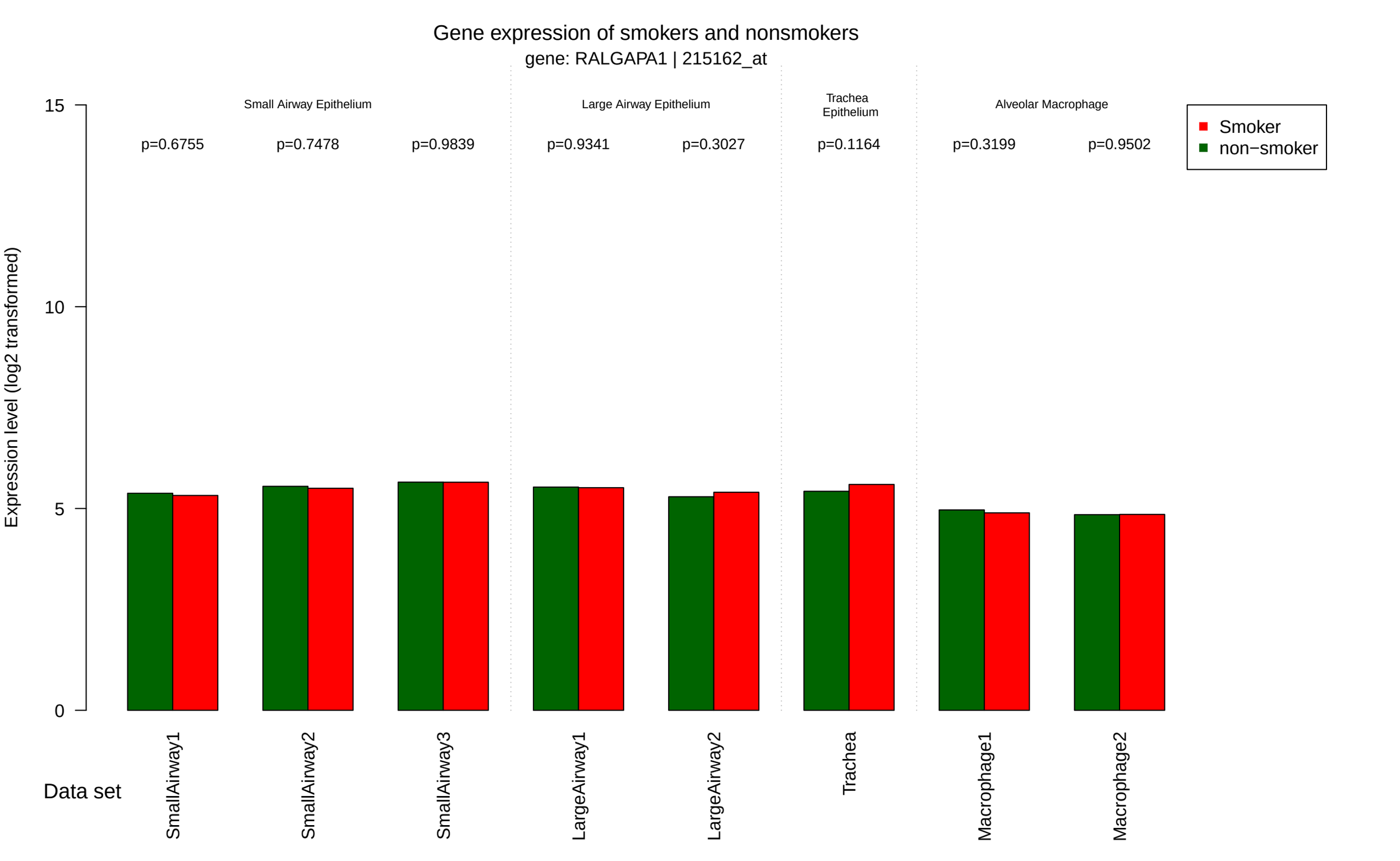
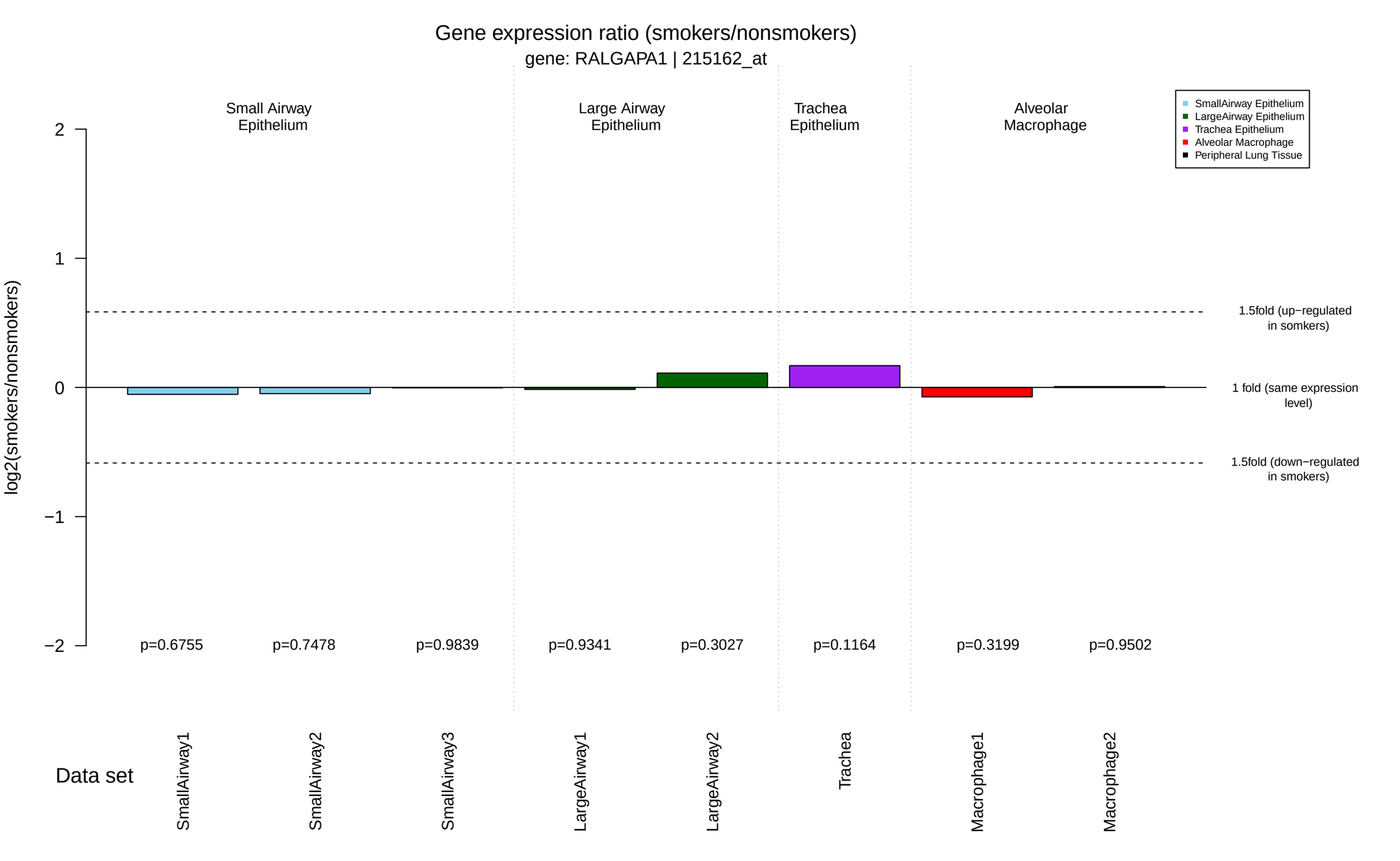
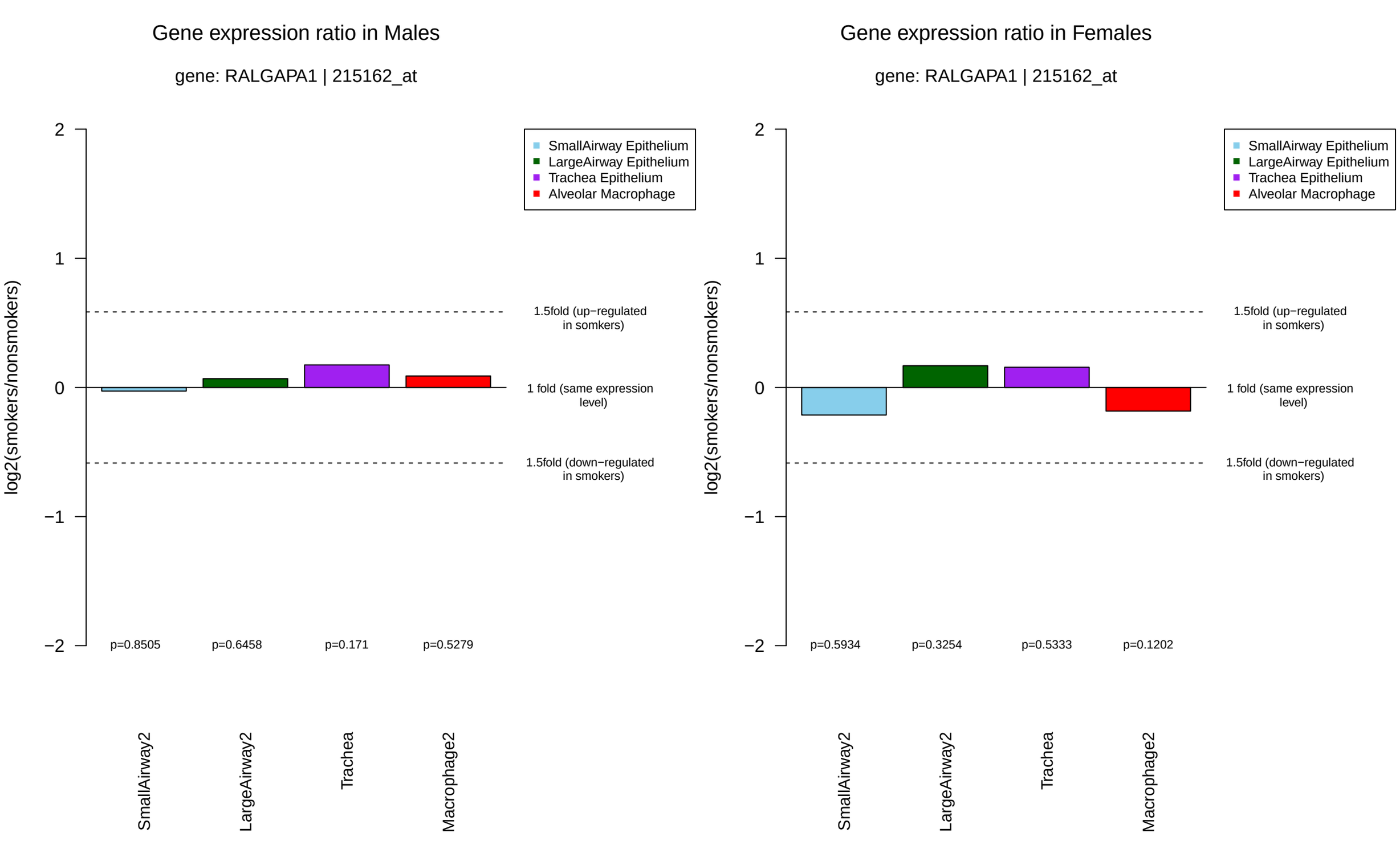
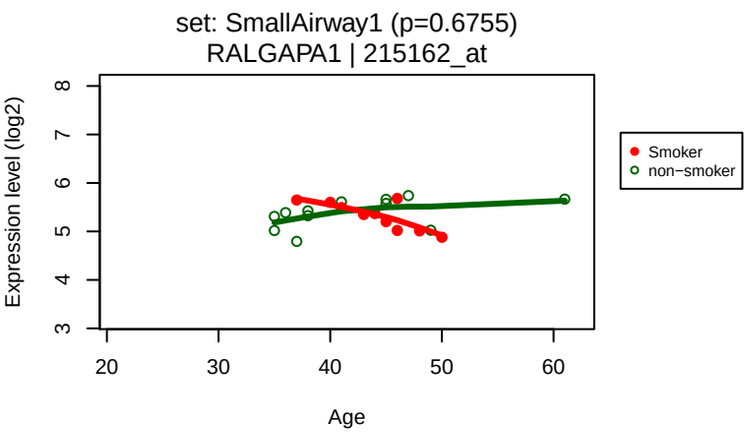
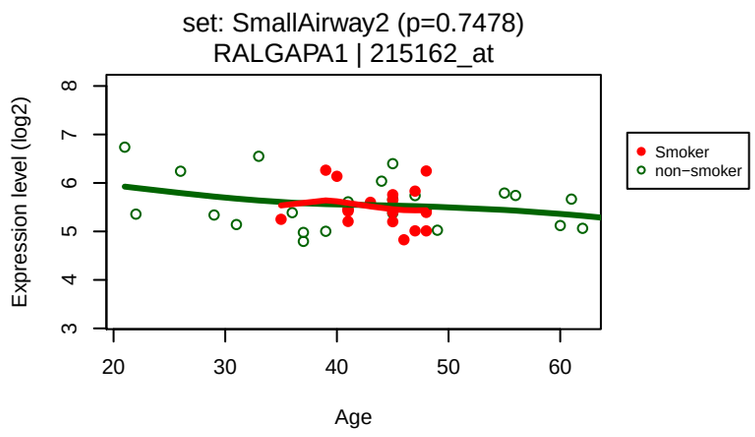
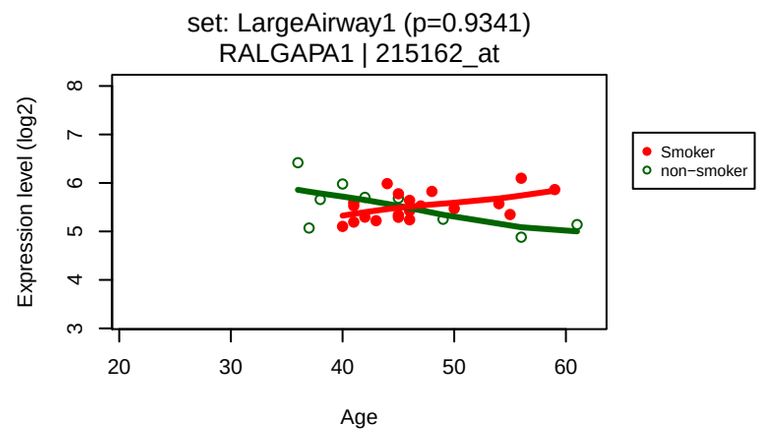

Gene expression of smokers and nonsmokers
gene: RALGAPA1 | 215162_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6755
p=0.7478
p=0.9839
p=0.9341
p=0.3027
p=0.1164
p=0.3199
p=0.9502
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

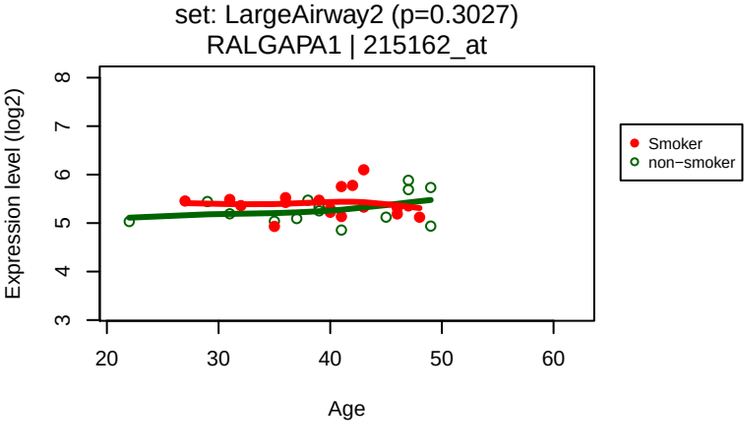

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA1 | 215162_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6755
p=0.7478
p=0.9839
p=0.9341
p=0.3027
p=0.1164
p=0.3199
p=0.9502
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

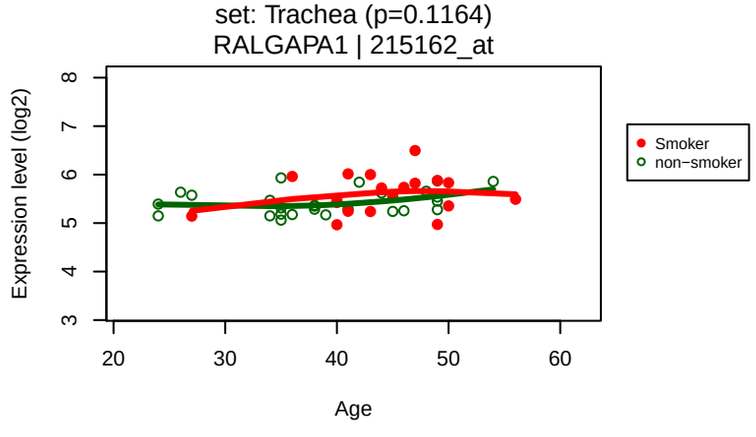

Gene expression ratio in Males
gene: RALGAPA1 | 215162_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8505
p=0.6458
p=0.171
p=0.5279
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA1 | 215162_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5934
p=0.3254
p=0.5333
p=0.1202
SmallAirway2
LargeAirway2
Trachea
Macrophage2

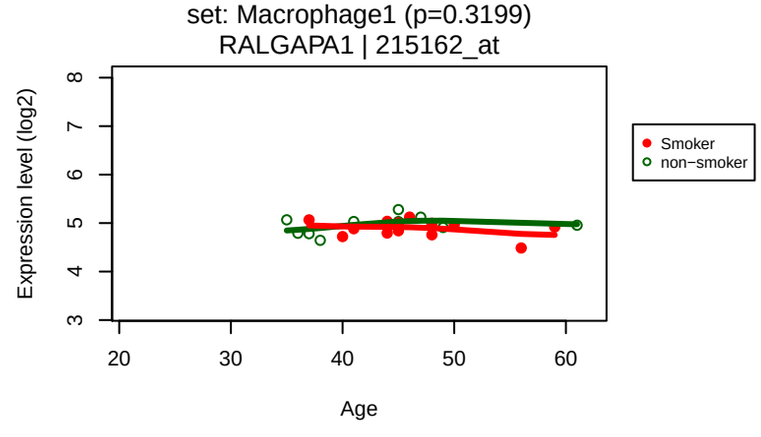

set: SmallAirway1 (p=0.6755)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker

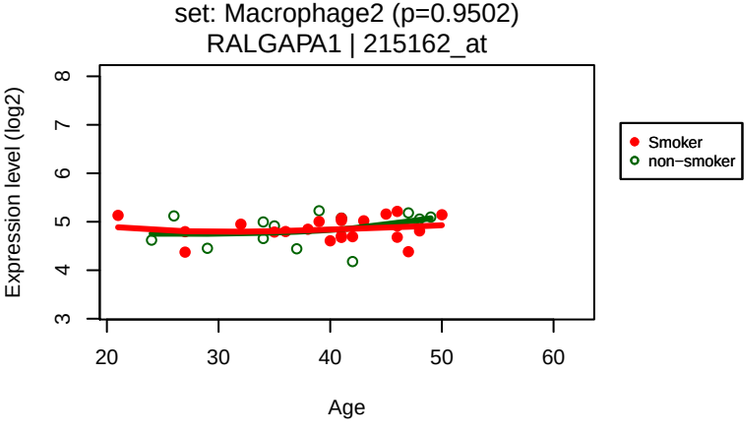

set: SmallAirway2 (p=0.7478)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker

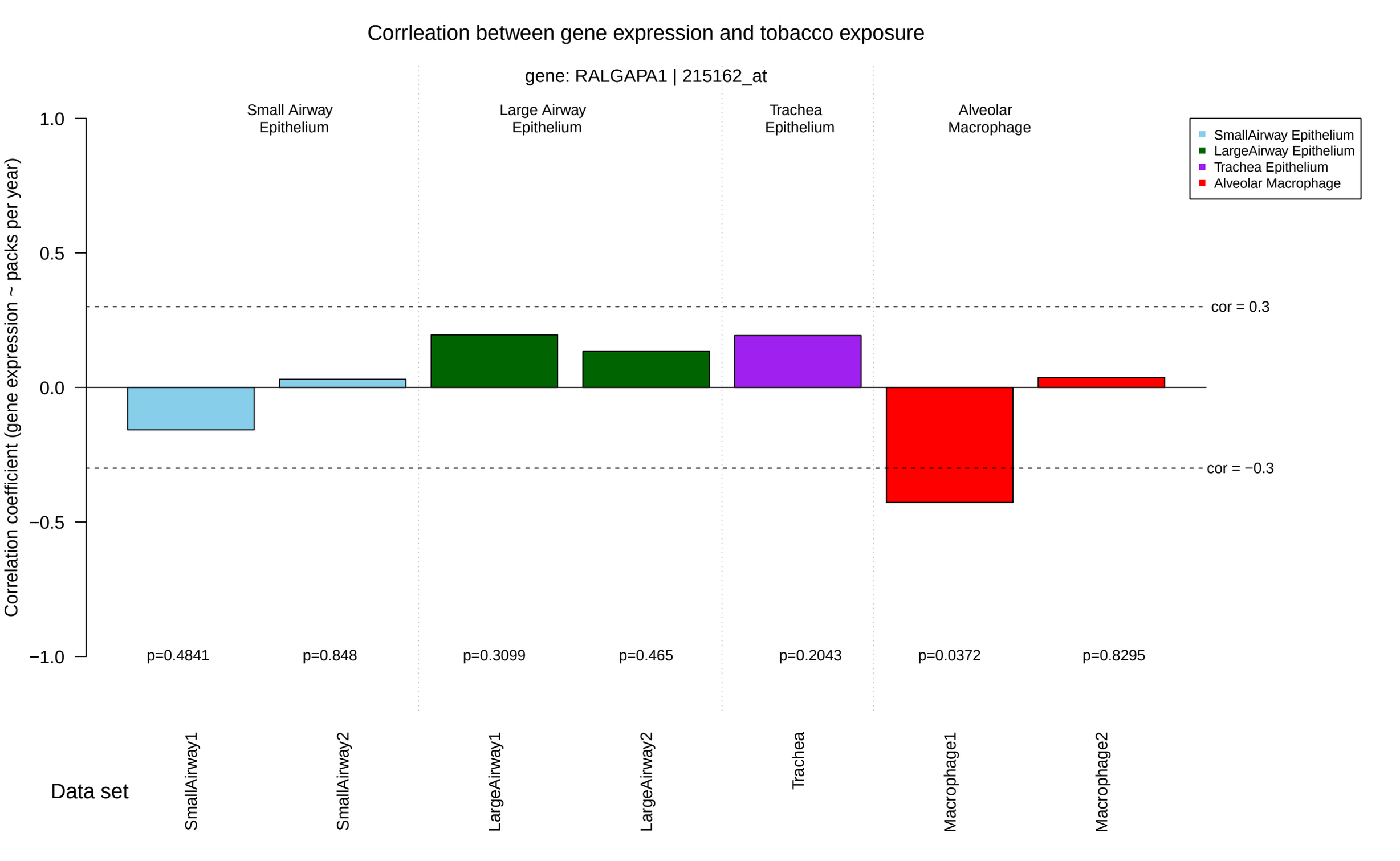

set: LargeAirway1 (p=0.9341)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3027)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1164)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3199)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9502)
RALGAPA1 | 215162_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA1 | 215162_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4841
p=0.848
p=0.3099
p=0.465
p=0.2043
p=0.0372
p=0.8295
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2